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MILITARY SPECIFICATION

RECEIVING SET, RADIO, AN/URR-35( )

1. SCOPE

1.1 This specification covers the design and performance of a UHF/VHF communications receiving set designed for ship, shore and mobile installations. The receiver is capable of being set to one preset crystal controlled channel or manually tuned over its entire frequency range.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-E-1 - Electron Tubes.
JAN-P-642 - Plugs, Telephone.
MIL-C-915 - Cable, Cord, and Wire, Electrical.
MIL-C-3098 - Crystal Units, Quartz, (Shipboard Use).
MIL-C-3608 - Connectors, "BNC" for Radio Frequency Cables.
MIL-I-15024 - Identification Plates, Information Plates, and Marking Information for Identification of Electric, Electronic, and Mechanical Equipment.
MIL-E-15090 - Enamel Equipment, Light Gray (Formula No. 111).
MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-I-16910 - Interference Measurements, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.
MIL-T-17113 - Tests, Shock, Vibration, and Inclination (for Electronic Equipment) (General Specification).
MIL-E-17555 - Electronic Equipment and Associated Maintenance Parts, Preservation, Packaging, Packing and Marking of.
MIL-P-17652 - Preparation for Delivery of Electronic, Electrical and Electro-Mechanical Parts.
MIL-M-17952 - Modification Kit for Receiving Sets, Radio AN/URR-13, AN/URR-27, and AN/URR-35.

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NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-108 - Definitions of and Basic Requirements for, Enclosures for.

DRAWINGS

BUREAU OF SHIPS RADIO
RE10F479 - Control Knobs for Electronic Equipment.
RE23F225 - Standard 19-Inch Relay Rack and Panel Dimensions.
RE49AA399 - Cap, Resistor, MX-554/U, Type BNC.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS-91906 - Instruction Book for Radio Receiving Set AN/URR-35.
NAVSHIPS-92022 - Instruction Book for Radio Receiving Set AN/URR-35A.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Material entering into the construction of each equipment shall be in accordance with Specification MIL-E-16400, except as otherwise specified herein.

3.1.1 Special alloys.- The use of "invar" or similar alloys for applications requiring materials having low temperature coefficients of expansion, and special steel alloys for audio or power frequency transformer or inductor cores shall be acceptable.

3.1.2 Screws and bolts.- Wherever practical, machine screws and bolts used throughout the set assembly shall employ "Phillips" head, or equivalent, screw driver slots.

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3.2 General.- The equipment shall meet the requirements of Specification MIL-E-16400, except as otherwise specified herein. The design objectives, as specified in Specification MIL-E-16400 for reliability, simplicity, ease of installation and maintenance consistent with the needs of the Naval service, shall be considered during all phases of the equipment design.

3.2.1 Temperature.- The equipment shall meet the temperature requirements of class 4 of Specification MIL-E-16400.

3.2.2 Control knobs.- Control knobs shall be in accordance with Drawing RE10F479.

3.2.3 Fuses.- Fuses shall be in accordance with Specification MIL-E-16400 and shall be contained in insulated extractor type fuse holders.

3.2.4 Blown fuse indicators.- Blown fuse indicators will not be required.

3.2.5 Enclosure.- The degree of enclosure shall be enclosed ventilated.

3.2.5.1 The set shall contain modification kit No. 1 for AN/URR-35 as described in Specification MIL-M-17952.

3.2.6 Time meters.- Time meters will not be required.

3.2.7 Finish.- The final finish ~~coats of~~ external surfaces shall be gray enamel conforming to type III, class 2 of Specification MIL-E-15090.

3.3 General characteristics.- The Radio Receiving Set AN/URR-35( ) shall comply with the following:

(a) Frequency range.- 225 to 400 megacycles (mc.), nominal. This range shall be continuous and without frequency hiatus.
(b) Frequency bands.- One.
(c) Number of preset channels.- One.
(d) Type of frequency control.- Crystal for preset channel; self-excited for continuous manual tuning.
(e) Type of receiver.- Double superheterodyne.
(f) Intermediate frequencies.- 18.6 mc. ± 2 kilocycles (kc.) and 1.775 mc. ±3 kc.
(g) Receiver outputs.-
Audio output connector.- 60 milliwatts into 600 ohms or 600 milliwatts into 600 ohms.
Phone jack.- 60 milliwatts into 600 ohms.
Scan output connector.- 10 microvolts across 50 ohms for input signal level of 20 microvolts.

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(h) Type of reception. - Amplitude modulated signals.
(i) Crystals. -
First oscillator. - JAN-type CR-24/U in accordance with Specification MIL-C-3098 (not to be supplied as a part of equipment furnished by contractor).
Second oscillator. - JAN-type CR-23/U in accordance with Specification MIL-C-3098 (shall be furnished as an integral part of the equipment).
(j) Adjacent channels spacing. - 100 kc., minimum.
(k) Input. - 50 ohms, nominal, unbalanced to ground.
(ℓ) Power source. - 105/115/125 volts, 50-60 cycles, single phase. Maximum power 120 watts at 115 volts.
(m) Overall dimensions. -
Height (excluding shock mountings) 6-31/32 ± 1/16 inches.
(excluding shock mountings) 8-7/16 ± 1/8 inches.
Width (including relay mounting brackets) 17-1/2 ± 0-1/16 inches.
(including relay mounting brackets) 19 ± 1/16 inches.
Depth (overall) 19-1/8 ± 1/16 inches.
(front of relay mounting bracket) 2-5/16 ± 1/16 inches.
(n) Weight. - 60 pounds maximum with shock mounting excluding relay mounting brackets.

3.4 Definitions. - The definitions herein are applicable to the equipment covered by this specification. In all tests conducted to determine specification conformance, cognizance shall be taken of these definitions as pertinent to the tests.

3.4.1 Standard dummy antenna. - A standard dummy antenna shall be a device simulating the impedance characteristics of an antenna for use with the equipment throughout its frequency range. A 50-ohm noninductive resistance shall be employed when it is required that radio frequency (r.f.) inputs be applied to the input circuits from a signal generator. This resistance, intended to simulate the nominal average impedance of the concentric transmission line, shall include the resistance arm of the output attenuator in shunt with the output terminals of the standard signal generator.

3.4.2 Standard signal to noise. - The signal-to-noise ratio shall be the ratio between a standard audio output signal and the output due to noise when the modulation is removed from the r.f. carrier. Standard signal-to-noise ratio shall be 10 decibels (db.).

3.4.3 Standard audio output. - Standard audio output shall be 6 milliwatts with respect to a 600-ohm noninductive resistance load across the output terminals of the receiver.

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3.4.4 Standard scanning output.- Standard scanning output shall be 10 microvolts with respect to a 50-ohm noninductive load across the scanning output terminals of the receiver.

3.4.5 Standard gain.- Standard gain shall be that adjustment of the receiver gain that produces standard signal-to-noise ratio.

3.4.6 Standard modulation.- Standard modulation shall be 30 percent at 1000 cycles per second (c.p.s.).

3.4.7 Sensitivity.- Sensitivity of a radio receiver is that characteristic which determines the minimum strength of signal input capable of causing a desired value of signal output.

3.4.8 Standard audio sensitivity.- Standard audio sensitivity shall be the number of microvolts of standard modulated signal required to produce standard audio output when applied at the receiver input through a standard dummy antenna with the receiver adjusted for standard gain. Microvolts shall be the open circuit voltage measured at the generator or dummy antenna terminals.

3.4.9 Standard scanning sensitivity.- Standard scanning sensitivity shall be the number of microvolts of unmodulated signal required to produce standard scanning output when applied at the receiver input through a standard dummy antenna. Microvolts shall be the open circuit voltage.

3.4.10 Selectivity.- Selectivity of a radio receiver is that characteristic which determines the extent to which it is capable of differentiating between the desired resonant signal and radio frequency signals of other frequencies.

3.4.11 Weak signal selectivity.- Weak signal selectivity shall be the degree to which the equipment is capable of differentiating between the desired signal and relatively weak interfering signals of frequencies relatively close to the desired signal frequency. It shall be expressed in terms of the attenuation, in decibels, the equipment introduces to frequencies removed from resonance with respect to the resonant sensitivity, using standard output as a common reference level.

3.4.12 Cross modulation.- Cross modulation rejection shall be the degree to which the receiver is capable of preventing strong modulated interfering signals relatively close to the desired signal frequency from modulating the carrier of the desired signal. It shall be expressed in terms of the attenuation, in decibels, the receiver introduces to 30 and 90 percent modulated interfering signals, at various frequencies off resonance, with respect to the resonant sensitivity, when using a 6 db increase in output due to the interfering signal modulation over the noise level with no interfering signal as a reference output.

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3.4.13 Intermodulation.- Intermodulation rejection shall be the degree to which the receiver is capable of rejecting two undesired signals whose frequencies are such that, after mixing in the receiver, a signal results which has the same frequency as the desired signal to produce an output signal.

3.4.14 Desensitization.- Desensitization shall be the reduction in sensitivity of the set as a result of an interfering signal at any frequency outside the channel to which the receiver is tuned. It shall be expressed in terms of the reduction in output to the desired signal as a result of the undesired signal.

3.4.15 Image rejection.- Image rejection shall be the degree to which the receiver is capable of rejecting signals which are removed from resonance by twice the frequency of the intermediate frequency amplifier. It shall be expressed in terms of the attenuation, in decibels, the receiver introduces to the image frequency with respect to resonant sensitivity, using standard output as a common reference level. Image rejection shall be expressed relative to the resonant frequency rather than the image frequency.

3.4.16 Intermediate frequency rejection.- Intermediate frequency rejection shall be the degree to which the receiver is capable of rejecting signals having the frequency of the intermediate frequency amplifier. It shall be expressed in terms of the attenuation, in decibels, the receiver introduces to such frequencies with respect to the resonant sensitivity, using standard output as a common reference level.

3.4.17 Spurious response rejection.- Spurious response rejection shall be the degree to which the receiver is capable of rejecting signal off resonance which can beat with the fundamental or harmonics of the oscillator or multipliers to produce the frequency of the intermediate frequency amplifier. It shall be expressed in terms of the attenuation, in decibels, the receiver introduces to spurious signals with respect to resonant sensitivity, using standard output as a common reference level.

3.4.18 Maximum undistorted output.- Maximum undistorted output shall be the output for increasing values of standard modulated input at which the total harmonic content is 7 percent with the receiver adjusted for standard gain.

3.4.19 Threshold of silencer.- Threshold of operation of the silencer shall be the radio frequency input level at which an audio output signal is produced, but below this level the receiver is silenced.

3.4.20 Threshold of a.g.c.- The threshold of the automatic gain control shall be the radio frequency level at which the overall gain is decreased by 1 db.

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3.5 Circuit description.- Radio Receiving Set AN/URR-35( ) shall be a double conversion superheterodyne receiver capable of receiving amplitude modulated transmissions in the very high and ultra high frequency region. The resonant signal frequency shall be fed from the antenna connector to the input of the first mixer through two stages of selective tuned radio frequency amplification. The local signal to the first mixer shall be obtained from an oscillator whose output is fed through three stages of frequency multiplication consisting of two doublers and a tripler. The first local oscillator shall be capable of operating crystal controlled, for single channel operation, or self-excited for continuously variable manual tuning. The r.f. amplifiers and the first oscillator, including the associated multipliers, shall be ganged together so as to be tunable over the complete range by a single tuning control for both crystal control and manual tuning operation. The output of the first mixer shall be followed by a single stage of selective amplification at a nominal frequency of 18.6 mc. and feed to the input of a second converter where it shall be mixed with the output of a second fixed frequency, crystal controlled oscillator to produce the final intermediate frequency having a nominal frequency of 1.775 mc. Two stages of selective amplification at 1.1775 mc. shall be employed between the output of the second mixer and the input of the first detector. Three stages of audio amplification shall be employed between the final detector and output connections. The receiver shall also incorporate circuits and facilities, (1) for broad band output at the frequency of the first i.f. amplifier for connection of a panoramic adaptor; (2) peak amplitude noise limiting; (3) automatic control of the r.f. gain, and (4) silencing the output to noise in the absence of a signal carrier. All a.c. and d.c. potentials necessary for the operation of the set shall be obtained from a self-contained power supply. Filters shall be employed between the power and audio connectors and their associated circuits to reduce the possiblity of interference entering the set from the interconnecting cables.

3.6 Type of construction.- The equipment shall be of the cabinet type of construction for both relay rack and top of table mounting. The cabinet shall be a five sided enclosure with the panel of the chassis-panel assembly forming the front and sixth side of the equipment. All corners, edges and projections of the cabinet and panel shall be rounded to a radius of not less than 1/4 nor more than 1/2 inch. The edges of the front panel shall be lipped so as to lap as inner protruding flange on the front edges of the cabinet in such a manner that the outer mating surfaces of the cabinet and panel are flush. The circuit parts shall be grouped, on a functional basis, into five major sections; namely, a preselector, intermediate and audio amplifiers, power supply, cable filters, and front panel. The first three sections shall be assembled within the chassis frame with the front panel section attached to the front of the frame. The cable filtering section shall be mounted against the rear wall of the cabinet. The chassis shall contain a thermostatically controlled blower to maintain the operating temperature of the set within safe limits. Intake and exhaust openings in the cabinet shall comply with the applicable requirements of Standard MIL-STD-108.

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3.7 Performance.- The equipment shall comply with the following performance requirements under the specified standard conditions for testing, unless otherwise specifically stated in section 4.

3.7.1 Sensitivity.- The sensitivity of the receiver shall not be poorer than 8 microvolts over the nominal tuning range for audio output. The scanning sensitivity shall not be poorer than 20 microvolts over the nominal tuning range.

3.7.2 Gain variation.- The variation in gain shall not exceed 10 db over the complete frequency range of the receiver.

3.7.3 Reserve gain.- In order to provide for changes in sensitivity of the receiver resulting from its operation under extreme conditions of temperature and humidity, reduction of power supply voltage, aging of tubes, circuit maladjustment, or similar items there shall be provided a reserve gain of not less than 12 db.

3.7.4 Weak signal selectivity.- The total bandwidth at 6 db attenuation shall not be greater than 85 kc, nor less than 70 kc. The total bandwidth at 60 db attenuation shall not exceed 190 kc. The weak signal selectivity characteristic shall be centered with respect to the specified intermediate frequency; and shall possess the maximum possible degree of symmetry for attenuation points up to 100 db with no evidence of double peaks at the nose, or irregularities in skirt configuration due to regeneration, electron tube overloading, oscillator pulling, or similar factors.

3.7.5 Scanning selectivity.- The selectivity of the receiver, measured from r.f. input to scanning channel output, shall have the following characteristics:

Attenuation db	Bandwidth kc.
6	300 minimum, 1000 maximum
20	2000 maximum

3.7.6 Cross modulation.- The design of the equipment shall be such as to comply with the following:

Db above resonant sensitivity	Percent off resonance
60	±0.05
80	±0.1
100	±0.15

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3.7.7 Intermodulation rejection.- The set shall afford not less than 50 db to interference due to intermodulation.

3.7.8 Desensitization.- The minimum level of the undesired signals required to produce a 2 db decrease in the desired signal shall be as follows:

Undesired level (db above resonant) sensitivity	Mc. off resonance
60	0.1
80	0.2

3.7.9 Image rejection.- The image rejection for crystal operation shall be not less than 75 db over the frequency range 225 to 350 mc. Over the frequency range 350 to 400 mc., the image rejection shall be not less than the value represented by a linear slope of 75 db at 350 mc. to 65 db at 400 mc.

3.7.9.1 Crystal operation.-

3.7.9.2 Self-excited operation.- The image rejection for self-excited operation shall be not less than 60 db over the nominal frequency range of the receiver.

3.7.10 Spurious response rejection.- Rejection of spurious responses shall be not less than 70 db in the specified frequency range of 225 to 400 mc. and not less than 60 db to all other frequencies in the range of 100 to 1000 mc.

3.7.11 Intermediate frequency rejection.- Intermediate frequency rejection shall be not less than 120 db.

3.7.12 Maximum undistorted output.- Maximum undistorted output shall be not less than 6 volts as measured across a standard output load.

3.7.13 Overall fidelity.- The overall fidelity of the equipment shall meet the following requirements. All references to attenuation are with respect to 1000 cycles as the frequency of 0 db attenuation:

Modulation frequency	Attenuation
100 cycles	Not less than 20 db.
350 to 3500 cycles	Within 6 db total variation.
7000 cycles	Not less than 10 db.

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3.7.14 Radiated and conducted noise.- The set shall be capable of meeting the radiated and conducted interference requirements of Specification MIL-I-16910. The scan output shall be deleted from this requirement when properly terminated by the dummy scan load (see 3.8.1).

3.7.15 R.f. filter characteristics.- The r.f. power line filters employed between the primary power supply line and the power unit shall, in combination with the specified power unit, present an attenuation of not less than 80 db to any radio frequency between the limits of 2 to 400 mc. The r.f. power line filter shall be balanced with respect to ground. The intermediate freqency output connection shall be provided with a buffer amplifier and shall present not less than 60 db attenuation to signals entering the set through the scan connector. The sensitivity of the receiving equipment to external terminals (other than those intended to operate at r.f. potentials) shall be not less than 80 db poorer than the sensitivity to signals fed through antenna terminals. The determination of output terminal sensitivity, as contradistinct from antenna sensitivity, shall be determined with the antenna input circuit closed through a standard dummy antenna which, together with the antenna input terminals, shall be completely shielded.

3.7.16 Accuracy of main tuning dial calibrations.- The calibrated frequency scale of the main tuning dial shall be accurate to ±0.5 percent at all points on the dial scale. This requirement shall be interpreted as meaning that, with the main tuning adjusted to a given dial setting, the resonant frequency of the tuned elements when operating with self-excited first oscillator, may be as much as 0.5 percent higher or 0.5 percent lower than the absolute frequency of the applied radio frequency carrier.

3.7.17 Tuning linearity.- The variation of frequency with angular rotation of the main drive shafts associated with the tuning elements of the preselection circuits shall be such that the scale markings shall be of a uniform increment for the complete frequency range.

3.7.18 Automatic gain control characteristics.- The automatic gain control shall operate in such manner as to hold the audio level constant, within a total tolerance of 3 db, for any variation of modulated input between 2 and 100,000 microvolts. This requirement predicates that the a.f. level control is adjusted for standard signal to noise. This operation shall hold for any frequency within the range of the receiver.

3.7.19 A.g.c. time constant.- The a.g.c. shall operate in such manner that the time of operation and recovery shall not exceed 0.2 second for input signals up to 10,000 microvolts or 0.5 second for input signals up to 500,000 microvolts. The a.g.c. shall be considered as having operated when, after application of an MCW square

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keyed input to the receiver, the output has reached a value 63 percent of the steady amplitude of the final value. The a.g.c. shall be considered as having recovered when, after removal of an MCW square keyed input from the receiver, the sensitivity has reached a value, as indicated by an increase in noise level, equal to 63 percent of the steady amplitude of the steady value. Start of time shall be based on the moment of application or removal of the keyed signal.

3.7.20 Noise limiter characteristics.- The noise limiter shall function to reduce the effects of noise without affecting the desired signals to a significant extent. It shall effectively limit or reduce voltage surges, or peaks, exceeding the output obtained with a 30 to 40 percent modulated signal. The limiter in-out insertion loss shall not exceed 3 db at any frequency within the range of the equipment.

3.7.21 Silencer characteristics.- The silencer circuits and its associated control shall be such that silencing is not effective for at least 15 percent of the control rotation but becomes effective within 60 percent of its rotation. The silencer shall be considered effective when the output is reduced 35 db below standard audio output. At 80 percent rotation, the silencer opening level shall be not less than 25 nor more than 75 microvolts. At 100 percent rotation, the silencer opening level shall be not less than 50 nor more than 300 microvolts. The silencer opening level shall be defined as the r.f. input level which provides an audio output 2 db down from the output obtained with the silencer out. When the silencer is set to just effectively silence noise with no signal input, the silencer shall open to within 6 db of the unsilenced output with a 30 percent modulated carrier of 1.5 microvolts or less.

3.7.22 Regeneration.- The interstage gain, r.f., i.f. and a.f. shall be so proportioned that there shall be no evidence of unintended oscillations or instability under any condition of operation of the receiver when employing electron tubes having maximum transconductance values specified in Specification MIL-E-1, and a supply voltage 10 percent higher than the highest nominal primary line voltage specified herein.

3.7.23 Frequency stability.- The overall frequency stability under any combination of the following conditions shall not exceed ±0.008 percent when operating crystal controlled and ±0.1 percent when operating as a continuously variable receiver. The stability requirements shall be determined by comparing the receiver resonant frequency, under normal ambient conditions, to the receiver resonant frequency under extremes of any service condition. Receiver resonant frequency is defined as midway between 6 db attenuation points on the receiver selectivity curve:

(a) Variation in line voltage ±10 percent of the nominal.
(b) Variation in power line frequency from 50 to 60 cycles.
(c) Variation in ambient temperature as specified in Specification MIL-E-16400 (see 3.2.1).
(d) Variation in relative humidity as specified in Specification MIL-E-16400.
(e) Vibration and shocks as specified in Specification MIL-T-17113 for light-weight equipment.

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3.7.23.1 Shift in band center with a 60 db change in gain shall not exceed 15 kc.

3.7.24 Degradation of sensitivity.- Audio sensitivity change shall not exceed the following:

(a) Change in sensitivity with temperature	3 db
(b) Change in sensitivity with humidity	3 db
(c) Decrease in sensitivity with change in line voltage	3 db

3.7.25 Shielding integrity.- The receiver shall be effectively shielded to permit its operation in strong fields of radiated radio freqency energy. The shielding shall be such that an r.f. field (unmodulated) of 3 volts per meter at any frequency up to and including ±100 kc. of resonant frequency shall result in a signal equal to or less than standard sensitivity (as contradistinct from actual sensitivity) being picked up by the resonant circuits of the receiver so as to result in the attainment of an input meter reading equal to that which had been precalibrated against a standard signal generator as being indicative of a signal of standard sensitivity. This measurement shall be made with the antenna terminated by a shielded dummy antenna.

3.7.26 Output hum level.- The output hum level, chargeable to the second detector and audio amplifiers, as measured with all controls adjusted as to produce maximum hum, shall not exceed 0.4 microwatts into a standard output load. With a 10 to 100,000 microvolts unmodulated input, the hum output shall not exceed a level of -40 db, referred to standard output with a 10,000 microvolt standard modulated input. The output hum level shall be measured as the total of the fundamental and all harmonics, such as would be indicated by a rectifier type a.c. voltmeter, or equivalent instrument.

3.7.27 Standing wave ratio.- The standing wave ratio shall not exceed 4 to 1 at any frequency within the tuning range of the set.

3.7.28 Inverse feedback.- Inverse feedback shall be incorporated between the audio output and the interstage amplifier to maintain low output distortion and an output signal that shall remain constant within 3 db for a 10 to 1 change in output load.

3.8 Equipment composition.- Each complete equipment furnished shall consist of the following:

(a) 1 set of electron tubes, rectifier, regulators and lamps.
(b) 1 second oscillator crystal.
(c) 1 set external cable fittings (receiver end) less cables.
(d) 1 set of test cables.

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(e) 1 each special alignment tool.
(f) 1 set of shock mountings.
(g) 1 set mounting brackets.
(h) 2 instruction books.
(i) 1 Dummy scan load.

3.8.1 Dummy scan load.- The dummy scan load shall meet the requirements of Drawing RE49AA399, except that the terminating resistor shall be a 50-ohm 1/2 watt resistor.

3.9 Detail requirements.-

3.9.1 Servicing facilities.- The receiving set shall contain the following facilities to assist in test and maintenance:

3.9.1.1 Service cables.- Interconnecting cables (coaxial or multiconductor) shall be furnished with each set to provide electrical interconnection between the chassis and the input filter and cable filter assembly. These test cables shall be fitted with appropriate mating plugs to engage the chassis mounted plugs and their associated filter mounted receptacles.

3.9.1.2 Alignment facilities.- All r.f. circuits, intermediate frequency (i.f.) circuits, oscillators and multipliers shall be provided with adjustable trimmers or equally effective means to permit accurate alignment of the individual circuits. The trimmers shall provide sufficient leeway of adjustment to compensate for the effects of normal aging of electron tubes and other parts. A switch shall be provided to connect the input meter to the grid return circuit of the tripler to assist in alignment of the first oscillator and associated multipliers.

3.9.1.3 Test connections.- Jacks and test connections, as appropriate, shall be provided to afford connection to not less than the following circuit positions. Connection of the test instruments shall not affect the normal circuit function. Precautionary measures, as necessary, shall be adequately covered by the instruction book:

(a) Input to the 1st doubler.
(b) Input to the 2nd doubler.
(c) Input to the tripler.
(d) Output of the 1st mixer (input to 1st i.f. amplifiers).
(e) Second oscillator grid.
(f) Output of the 2nd mixer (input to 2nd i.f. amplifiers).
(g) Output of the final detector (diode output).
(h) A.g.c. bus.
(i) High voltage d.c. bus.

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3.9.2 Cabinet construction.- The cabinet assembly shall be of sturdy design and construction compatible with the type of service for which it is intended. The cabinet shall be fabricated from aluminum or aluminum alloy. All seams shall be seam or spot welded so as to insure good electrical contact throughout their lengths. If spot welding is employed, such welding shall be accomplished by spacing the welds at not greater than 3/4-inch intervals. Additionally, the spot welding shall be so controlled as to assure uniformity of welding without undue formations of welds craters which would be detrimental to a smooth final surface finish. Guides or runners shall be provided on the interior of the cabinet to permit easy removal of the chassis from the cabinet.

3.9.2.1 Rack mounting facilities.- Rack mounting facilities shall be provided by means of removable brackets which shall be furnished as parts of each equipment. The mounting brackets shall be attached to the sides of the cabinet by means of screws or bolts which thread into reinforced members attached to the inside of the side walls of the cabinet assembly. The tapped holes for receiving the mounting screws or bolts shall be "blind", that is through tapped holes shall not be employed. The flanges of the brackets shall be notched as shown on Drawing RE23F225, for mounting on a standard relay rack. The flanges of the brackets shall be set flush with the flat portion of the side walls of the cabinet in such a manner that they do not project beyond the rounded edges of the front panel. The mounting brackets and the cabinet assembly shall be such as to provide all the necessary support for the chassis panel assembly when the equipment is mounted in a relay rack, without the use of additional brackets or reinforcing members to preclude the development of malformations either to the chassis panel assembly or to the cabinet assembly itself. The brackets shall be such as not to restrict the removal of the chassis from the front of the cabinet when installed in the relay rack.

3.9.2.2 Table (shipboard) mounting facilities.- Table mounting facilities shall be provided by the use of shock mountings which shall attach to the bottom surface of the cabinet by means of screws, nuts, and lockwashers, and shall be capable of being removed when it is desired to rack mount the set. The mountings shall be so placed that their centers will be 12-3/8 inches relative to the width of the set and 10-27/32 inches relative to the depth of the set.

3.9.3 Chassis-panel construction.- The chassis panel assembly and all brackets, shields and cover plates, and similar items, shall be fabricated from aluminum or aluminum alloy. The chassis shall be of unit construction wherein individual subassemblies are employed and secured together to form a composite assembly. The front operating panel shall be secured to the chassis by means of screws which shall engage mechanically self-locking inserts riveted to the chassis to form the chassis panel assembly. When the chassis is completely inserted into the cabinet, the chassis mounted plugs and associated filter receptacles shall automatically engage to circuitally connect the chassis with the rear mounted cable receptacles.

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3.9.3.1 Chassis detail.- The chassis assembly shall present a compartmentalized unit wherein all subassemblies are designed for economy of space utilization and minimum size and weight for the composite assembly. The chassis assembly shall be provided with complete shielding of all electrical parts mounted thereon, through the use of auxiliary shields, cover plates, and similar items as necessary to permit operation of the equipment when removed from its cabinet. Additionally, cover plates shall be provided for protection of the chassis mounted parts against damage from dirt or dust accumulations or damage which might otherwise be incurred during normal handling and installation of the equipment. Auxiliary shields, cover plates, and similar items shall be of such design that, when assembled and secured on the chassis, they shall present a pleasing appearance and provide positive electrical contact. The securing devices shall be of such design as to permit the attachment or detachment of the auxiliary shields, cover plates, and similar items by means of a conventional blade screwdriver or a small coin (dime or penny). The securing devices shall operate approximately 1/2 turn for complete tightening or loosening. Additionally they shall be captive with the shields, cover plates, and similar items, which they secure, and shall engage with mating locking inserts on the chassis or frame. DZUS fasteners (or equivalent) are intended by this requirement. The chassis shall be provided with stops functioning to retain the chassis when partially withdrawn from the cabinet sufficient to provide accessibility to tubes mounted on the i.f./a.f. chassis. The stops shall be equivalent of those employed on the Government furnished prototype Radio Receiving Set AN/URR-35. The design of the securing devices shall be such as to permit release of the chassis without aid of tools, and their operation shall be such as not to subject the fingers of an operator to possible injury. The chassis assembly shall be easily and completely removable from the front of the cabinet without disturbing the cabinet when mounted either on top of a table or on a standard relay rack.

3.9.3.1.1 Output transformer.- The secondary winding of the output transformer shall be electrostatically shielded and electromagnetically balanced, with respect to a grounded midpoint, so as to substantially remove all radio frequency potentials from the audio output and head-telephone terminals.

3.9.3.1.2 Power supply.- The rectifier power supply shall be an integral part of the receiver chassis assembly. It shall contain all electrical circuits and parts necessary for taking power from the specified a.c. power source and converting such power into low voltage a.c. power and rectified high voltage d.c. power to fulfill all a.c. and d.c. power requirements of the complete equipment. It shall include voltage regulator tubes as necessary to meet the requirements of this specification for frequency variations under conditions of variations in primary power supply voltage.

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3.9.3.1.2.1 Power inputs.- Taps shall be provided on the primary winding of the power transformer to permit operation from rated power sources of 105, 115 and 125 volts. Performance with voltage variations shall be predicated on a rated power source of 105 volts being applied to the 105-volt tap, or 115 volts being applied to the 115-volt tap, or 125 volts on the 125-volt tap. Selection of the appropriate primary tap shall be considered as an installation matter; that is, selection being made to suit the available power source. Connection to the selected tap shall be effected through the use of mechanical links. Switches, in the conventional sense, shall not be employed.

3.9.3.2 Panel detail.- The front panel mounted securing devices employed to retain the chassis in the cabinet shall be operable with a conventional blade screwdriver and shall operate with approximately 1/2 turn for complete tightening or loosening. The front flanges of the panel shall be provided with wiper springs to provide positive electrical bonding between the front panel and the cabinet assembly. The front panel shall be equipped on either side with handles to permit ease of inserting or removing the chassis-panel assembly into or out of the cabinet assembly without placing undue strain on the panel operating controls or other devices mounted thereon. Each handle shall be capable of supporting the full weight of the equipment.

3.9.4 Controls and devices.- Controls and devices shall be considered in two groups, one group designated as operating (continually required), and the other as semi-adjustable (infrequently required).

3.9.4.1 Operating controls.- The following operating controls and devices shall be those panel mounted items which may require frequent manipulation in normal operation of the equipment, or are provided for indication purposes. Controls and devices listed as (a) through (g) below shall be contained within a well recessed into the front panel. The well shall be of such depth as to limit the control knobs from extending more than 1/8 inch beyond the front of the panel:

(a) Main tuning control knob.
(b) Dial lock.
(c) Dimmer control.
(d) Silencer switch.
(e) Head-telephone level control.
(f) Power switch.
(g) Head-telephone jack.
(h) Main tuning dial.
(i) Input meter.
(j) Output meter.
(k) Neon lamp indicator.

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specifications or other data is not to be regarded by implication or otherwise as

3.9.4.1.1 Titles, markings and functions of panel controls and devices.- The panel controls and devices shall be titled, marked and shall operate as follows:

3.9.4.1.1.1 Main tuning control knob.- No title or markings will be required. This control knob shall contain a handle to permit ease in setting up the equipment to the desired frequency. Main tuning shall be accomplished by vernier control only; no direct drive shall be provided. The tuning drive shall be of such design as to permit easy rotation and rapid traversing of the tuning dial. The tuning drive shall have a ratio of not less than 15 to 1, nor more than 20 to 1.

3.9.4.1.1.2 Dial lock.- Title "LOCK". This device shall comply with the applicable requirements of Specification MIL-E-16400.

3.9.4.1.1.3 Dimmer control.- Title "DIMMER". This control shall function as specified in Specification MIL-E-16400.

3.9.4.1.1.4 Silencer switch.- Title "SILENCER". This shall be a toggle switch functioning to insert or release the silencer control circuits. It shall have its two positions marked "IN" and "OUT" with the "IN" applying in the upper position.

3.9.4.1.1.5 Head-telephone level control.- Title "PHONES". This control shall operate independent of the audio level control to adjust the level as the head-telephone jack over a range of not less than 40 db. It shall permit a continuous variation in output, or in steps not greater than 2 db, throughout its range. It shall provide a linear relationship, within a total variation of 6 db, within the limits of 0 to 40 db, between its angular rotation and the corresponding output (in decibels). Eleven division markings shall be equally spaced around the arc of operating range of the control. Every other division line shall be identified by the numerals "0-2-4-6-8-10" in left to right sequence.

3.9.4.1.1.6 Power switch.- Title "POWER". This shall be toggle switch functioning to connect and disconnect both legs of the primary power input. It shall have its two positions marked "ON" and "OFF" with the "ON" applying in the upper position.

3.9.4.1.1.7 Head-telephone jack.- No title or marking will be required. This jack shall mate with a standard JAN type PJ-055B head-telephone plug conforming to Specification JAN-P-642. It shall be provided with a hinged cover cap.

3.9.4.1.1.8 Main tuning dial.- No title will be required. The main tuning dial shall be as large as practical, preferably not less than 3 inches outside diameter. The dial shall be calibrated in frequency over not less than 240 angular degrees, such markings shall be made at each 2-megacycle increment with each fifth marking indexed

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scale. This requirement shall be interpreted as meaning that, with the main tuning

and each tenth marking carrying the appropriate frequency identification. The calibration shall indicate an increase in frequency with clockwise rotation. The orifice for viewing the dial scale shall be of such size as to enable an operator to see not less than 10 nor more than 17 dial increments at all times. The orifice shall contain a glass window protected against breakage due to shock and vibration. The dial shall be illuminated by means of two 6-volt pilot lamps mounted behind the operating panel.

3.9.4.1.1.9 Input meter.- Title "INPUT". This instrument shall be a 0 to 1 d.c. milliampere meter functioning primarily to indicate the change in current of the intermediate frequency amplifiers with change in a.g.c. voltage. (See 3.9.1.2 and 3.9.4.2.1.8.)

3.9.4.1.1.10 Output meter.- Title "OUTPUT". This instrument shall be a Navy type-22427 d.c. rectifier type meter with a -10/0/+20 db scale. (Information concerning the Navy type-22427 meter will be furnished to the contractor upon request to the bureau or agency concerned.) This meter shall function to indicate the power level delivered across a 600 ohm load.

3.9.4.1.1.11 Neon lamp indicator.- Title "CRYSTAL". This lamp shall function to glow when the equipment is adjusted for operation on crystal control. The lamp shall operate in conjunction with the oscillator switch so as to illuminate on crystal operation, and be extinguished when employing the self-excited oscillator. The lamp shall be energized by the rectified d.c. voltage from the power unit. The lamp shall be of the miniature type mounted so as to be accessible for removal through the front panel and protected by a white, translucent screw type bezel.

3.9.4.2 Semi-adjustable controls.- Semi-adjustable controls and devices shall be those (other than circuit alignment controls) which are normally preset to a fixed position or condition. The following controls and devices fall in this category and shall be contained behind access doors on either side of the operating panel. The locking devices for the doors shall be operable without the use of tools and shall operate with approximately 1/2 turn for complete tightening or loosening:

(a) A.f. level control.
(b) Silencer control.
(c) Noise limiter switch.
(d) Input meter adjustment.
(e) Fuses.
(f) Oscillator switch.
(g) Crystal holder.
(h) Alignment switch.

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3.9.4.2.1 Titles, markings and functions of semi-adjustable controls and devices.- The semi-adjustable controls and devices shall be titled, marked and shall operate as follows:

3.9.4.2.1.1 A.f. level control.- Title "A.F. LEVEL". This control shall be a screwdriver actuated potentiometer functioning to permit adjustment of the input to the audio frequency amplifiers over a range of not less than 40 db. It shall permit a continuous variation in output, or in steps not greater than 2 db, throughout its range. It shall provide a linear relationship, within a total variation of 6 db, within the limits of 0 to 40 db, between its angular rotation and the corresponding output (in decibels).

3.9.4.2.1.2 Silencer control.- Title "SILENCER". This control shall be a screwdriver actuated potentiometer functioning to permit adjustment of the threshold level of the silencer circuits. The silencer control shall have such a range as to permit adjustment of the threshold of the silencer circuit in steps not greater than 2 db over the range of 1.5 microvolts up to a maximum signal level of 300 microvolts, with 100 microvolts being the nominal level.

3.9.4.2.1.3 Noise limiter switch.- Title "N. L.". This shall be a toggle switch functioning to permit operation of the equipment with or without the noise peak limiter. It shall have its two positions marked "IN" and "OUT" with "IN" applying in the up position.

3.9.4.2.1.4 Input meter adjustment.- Title "INP. MTR". This control shall be a screwdriver actuated potentiometer functioning to permit adjustment of the input meter to its zero scale.

3.9.4.2.1.5 Fuses.- No title will be required. The fuses shall be contained in extractor type fuse holders.

3.9.4.2.1.6 Oscillator switch.- No title will be required. This control shall be a rotary switch functioning to permit selection between the crystal oscillator and the self-excited oscillator. Its two positions shall be marked "CRYSTAL" and "MANUAL" in left to right sequence.

3.9.4.2.1.7 Crystal holder (for 1st oscillator).- Title "CRYSTAL". This holder shall be capable of holding and maintaining contact with CR-24/U crystals under conditions of shock and vibration.

3.9.4.2.1.8 Alignment switch.- No title will be required. This control shall be a toggle switch functioning to permit precise adjustment of the main tuning control without the use of an external signal generator. The circuits, connected and removed by this switch, shall operate by applying a portion of the negative grid-bias voltage developed across the final multiplier grid-leak resistor to bias the a.g.c. bus in the receiver, thereby causing the input meter to indicate.

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3.9.5 Input cable and filter assembly construction.- All external interconnecting cables to the equipment shall terminate in receptacles contained on the bottom surface of an enclosure which shall extend not greater than 1-3/4 inches beyond the rear of the cabinet proper. All incoming and outgoing leads (exclusive of the antenna) shall contain radio frequency filter parts as necessary to provide attenuation to interfering signals picked up by the incoming cables. The filter assembly shall consist of a completely enclosed entity. All corners and edges of this enclosure, except the bottom edge, shall be rounded to conform with the contours of the cabinet assembly. The bottom surface of the filter subassembly shall be placed at such an angle as to provide ample clearance for the attachment or detachment of cable plugs without resorting to the use of angle type fittings. The complete filter assembly shall be removable from and thorugh the front of the cabinet. The detachment of external cables will be permitted to effect this removal. The cover of the assembly shall be removable from the rear of the equipment without the necessity of disconnecting the incoming cables. Removing the cover shall provide access to all parts contained within the enclosure.

3.9.5.1 Cable receptacles.- The following receptacles shall be contained on the input cable and filter assembly, and mating plugs shall be furnished as parts of each complete assembly:

(a) A.c. power input	Title "A.C. POWER".
(b) Scanning input	Title "50 OHM SCAN".
(c) Antenna input	Title "50 OHM ANT.".
(d) Audio output	Title "AUDIO".
(e) Ground terminal	Title "GND.".

3.9.5.2 Cables.- The cable employed for power connection shall be in accordance with Specification MIL-C-915. Cables will not be furnished by the contractor.

3.9.5.3 Scan output.- The scan output receptacle shall be a UG-291/U in accordance with Specification MIL-C-3608.

3.10 Supplementary nameplates.- Supplementary nameplates shall be type A or B, aluminum alloy, size 1, identification plates in accordance with Specification MIL-I-15024, except that stock number, manufacturer's data, contract number, U.S. property and operating characteristics shall not be included. Figure 2 shall be used as a guide only. These identification nameplates shall be in addition to the nameplates required by the contract or order. One nameplate shall be mounted on the right hand front panel access door and one mounted on the inside bottom front of the cabinet.

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3.11 Government furnished prototype. - The design, appearance and circuitry of the AN/URR-35( ) shall be identical to the Government furnished prototype model AN/URR-35, as described in Publication NAVSHIPS 92022 or 91906 without waiver or any obligation to effect full compliance with all the requirements specified herein for performance, construction and selection of parts. In addition, all parts and units employed in their construction shall be electrically and mechanically interchangeable part for part and unit for unit with corresponding parts and units of the Government furnished prototype model.

3.12 Workmanship. - Workmanship shall be in accordance with Specification MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - Sampling, inspection, and tests shall be as specified in Specification MIL-E-16400, except as otherwise specified herein.

4.2 Acceptability tests. - Acceptability tests shall be made on the prototype models at a Government laboratory or at place of manufacture as specified (see 6.1). In addition to the acceptability tests specified in Specification MIL-E-16400, tests shall be made to determine the following:

Test	Reference paragraphs
Power requirements	3.3 (ℓ)
Frequency range	3.3 (a)
Tuning ratio	3.9.4.1.1.1
Tuning linearity	3.7.17
Accuracy of calibrations	3.7.16
R.f. input impedance	3.3 (k)
Audio output impedance	3.3 (g)
Head-telephone level control characteristics	3.9.4.1.1.5
A.f. level control characteristics	3.9.4.2.1.1
Silencer characteristics	3.9.4.2.1.2 and 3.7.2.1
Noise limiter characteristics	3.7.20
A.g.c. characteristics	3.7.18 and 3.7.19
Sensitivity	3.7.1
Reserve gain	3.7.3
Gain variation	3.8.2
Weak signal selectivity	3.3 (j) and 3.7.4
Scanning selectivity	3.7.5
Image rejection	3.7.9
I.f. rejection	3.7.11
Spurious response rejection	3.7.10

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Test	Reference paragraphs
Cross modulation	3.7.6
Intermodulation rejection	3.7.7
Desensitization	3.7.8
Shielding integrity	3.7.25
Maximum undistorted output	3.7.12
Overall fidelity	3.7.13
Output hum level	3.7.26
Regeneration	3.7.22
R.f. filter characteristics	3.7.15
Frequency stability	3.7.23
Degradation of sensitivity	3.7.24
Standing wave ratio	3.7.27
Output with varying loads	3.7.28

4.3 Government inspection and tests. -

4.3.1 Production tests. - In addition to the production tests specified in Specification MIL-E-16400, tests shall be made on each production unit offered for delivery to determine compliance with the following:

Test	Reference paragraphs
Sensitivity (audio output only)	3.7.1
Reserve gain	3.7.3
Gain variation	3.7.2
Weak signal selectivity	3.7.4
Image rejection	3.7.9
I.f. rejection	3.7.11
Output hum (listening test)	-----
Control operation (air test)	-----

4.3.2 Production control tests. - Production control tests shall be made on one equipment from each 50 offered for delivery, except that the undesired radiation test and the radio interference will be made on one equipment from each 100 offered for delivery. It shall be at the discretion of the Government inspector to increase or decrease the number of units to be tested to assure proper production control. The tests shall be in accordance with Specification MIL-E-16400 and the following:

Test	Reference paragraphs
Accuracy of calibrations	3.7.16
A.g.c. characteristics	3.7.18
Spurious response rejection	3.7.10

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Test	Reference paragraphs
Maximum undistorted output	3.7.12
Overall fidelity	3.7.13
Regeneration	3.7.22

4.3.3 Environmental tests. - Environmental tests shall be made on one equipment from each 500 offered for delivery.

4.4 Standard conditions of tests. - Unless otherwise specifically stated in the contract or order, test measurements shall be made under the following standard conditions:

(a) Power supply input - 115 volts, 60 cycles, single phase a.c.
(b) R.f. input - Through a standard dummy antenna.
(c) A.f. output - Connected to a standard dummy load.
(d) Audio gain - Adjusted for standard gain.
(e) Noise limiter - Off.
(f) Scan output - Terminated by dummy scan load.
(g) Silencer - Off.
(h) Ambient temperature - Between 20° and 30° C.
(i) Relative humidity - Below 50 percent.
(j) Electron tubes - All electron tubes (except rectifiers, regulators and diodes) shall have transconductance (GM) values approximately equal to the mean value of the limiting figures specified in Specification MIL-E-1.

4.5 Test procedures. - The test set up and methods of tests shall be as specified hereinafter.

4.5.1 Audio channel sensitivity. -

4.5.1.1 Test set up. -

(a) Connection input signal generator, capable of generating frequencies in the range 225 to 400 mc., to the input of the AN/URR-35( ) through a standard dummy antenna (see 3.4.1).and a 3-foot section of 52-ohm coaxial cable.
(b) Connect an i.f. signal generator, capable of generating 18.6 mc., to test jack in the output circuit of the first mixer of the AN/URR-35( ).
(c) Connect a d.c. electron tube volt meter between the a.g.c. test jack and ground.
(d) Connect a voltmeter (Ballantine Model 300, or equivalent) across the standard output load (see 3.4.3) at the rear audio output connector of the AN/URR-35( ).

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4.5.1.2 Method of test.-

(a) Adjust the AN/URR-35( ) to the frequency at which test will be made.
(b) Adjust the input signal generator, (unmodulated) to the resonant frequency of the AN/URR-35( ) as indicated by maximum a.g.c. voltage with a weak signal and then reduce the output of the input signal generator to zero.
(c) Adjust the i.f. signal generator to 18.6 mc. and adjust output of the i.f. signal generator and the audio level control of the AN/URR-35( ) so that when modulation of 1000 cycles per second at 30 percent is on the output voltmeter reads 1.9 volts and with the modulation removed the output voltmeter reads 0.6 volt (see 3.4.2).
(d) Record the a.g.c. voltage level, as indicated by the d.c. electron tube voltmeter, and the i.f. signal generator output which occurs in step (c), then reduce output of the i.f. signal generator to zero.
(e) Increase the input generator until the a.g.c. voltage recorded in step (d) is obtained. Record the open circuit voltage as measured at the terminals of the signal generator or dummy antenna.

4.5.2 Reserve gain.-

4.5.2.1 Test set up.- The test set up shall be as specified for audio channel sensitivity in 4.5.1.1.

4.5.2.2 Method of test.-

(a) Same as (a) through (e) of 4.5.1.2.
(b) Record reading of audio output voltmeter obtained in step (e) of 4.5.1.2 as the standard output noise voltage.
(c) Adjust audio level control of the AN/URR-35( ) full clockwise and record reading of the audio output voltmeter as the maximum output noise voltage.
(d) Determine reserve gain by the following formula:

$$\text{Reserve gain} = \frac{\text{Specified sensitivity}}{\text{Actual sensitivity}} \times \text{Actual reserve gain ratio}$$

Legend: Specified sensitivity = 8 microvolts (see 3.7.1).
Actual sensitivity is value recorded in step (e) of 4.5.1.2.
Actual reserve gain ratio is the measured numerical ratio of the noise voltages recorded in steps (b) and (c) above.

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4.5.3 Scanning channel sensitivity.-

4.5.3.1 Test set up.-

(a) Same as (a) of 4.5.1.1.
(b) Connect the scan output of the AN/URR-35( ) to the antenna input of a test receiver, capable of resonating at 18.6 mc., through a 3-foot section of 52-ohm coaxial cable and a 0.01 mfd. blocking capacitor and connect a 50-ohm carbon resistor from the antenna connector of the test receiver to ground.
(c) Connect a d.c. electron tube voltmeter to the diode load of the test receiver.
(d) Install a crystal in the first oscillator of the AN/URR-35( ) and set crystal-manual switch for crystal operation.
(e) Disable a.g.c. of AN/URR-35( ).
(f) Connect a d.c. electron tube voltmeter to the detector diode test jack of the AN/URR-35( ).

4.5.3.2 Method of test.-

(a) Tune AN/URR-35( ) for maximum noise reading on the output meter and lock tuning dial.
(b) Set output of input generator to approximately 100 microvolts and tune generator, with modulation off, for maximum reading on the AN/URR-35( ) input meter.
(c) Tune test receiver for its maximum diode voltage reading.
(d) Disconnect test receiver from AN/URR-35( ) and connect a signal generator, capable of generating 18.6 m.c. to the input of the test receiver, and adjust to the resonant frequency of the test receiver as indicated by maximum diode voltage output.
(e) Record diode output of test receiver obtained with a 10 microvolt input.
(f) Connect test receiver to scan output of the AN/URR-35( ) as specified in (b) of 4.5.3.1.
(g) Adjust output of input signal generator (see 4.5.3.1 (a)) until diode reading recorded in (e) above is obtained. The generator output shall be recorded as the scanning channel sensitivity.

4.5.4 Intermediate frequency selectivity.-

4.5.4.1 Test set up.-

(a) Connect signal generator as specified in (b) of 4.5.1.1.

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4.5.4.2 Method of test.-

(a) Adjust generator output to 100 microvolts, modulated 30 percent at 1000 c.p.s., and tune generator for maximum output meter reading on the AN/URR-35( ).
(b) Adjust audio output level control of the AN/URR-35( ) to give 2 milliwatts output across a standard output load.
(c) Increase signal generator output to 2 times value of (a) and tune generator each side of resonance until 2 milliwatt output is again obtained. Measure and record the frequencies of the generator as the 6 db bandpass.
(d) Increase signal generator output to 1000 times the input of (a) and tune generator each side of resonance until 2 milliwatt output is again obtained. Measure and record the frequencies of the generator as the 60 db bandpass.

4.5.5 Intermodulation.-

4.5.5.1 Test set up.-

(a) Two input signal generators, capable of generating frequencies between 225 to 400 mc., shall be connected to a 50-ohm 6 db mixing pad through 3-foot lengths of 52-ohm coaxial cable.
(b) Connect third terminal of the 50-ohm 6 db mixing pad to the antenna input of the AN/URR-35( ).
(c) Same as (c) of 4.5.1.1.
(d) Same as (d) of 4.5.1.1.

4.5.5.2 Method of test.-

(a) Adjust one input signal generator, modulated 30 percent at 1000 cycles, to the resonant frequency of the AN/URR-35( ) and adjust generator output and the audio level control of the AN/URR-35( ) until standard signal plus noise-to-noise ratio is obtained. Record generator output.
(b) Record the a.g.c. voltage level obtained under condition which satisfies (a) above. Reduce output of this generator to zero.
(c) Adjust the second input generator (unmodulated) to the resonant frequency and set its output until the reference voltage obtained in step (b) above is obtained. Record generator output.
(d) Connect a frequency meter to the scan channel output of the AN/URR-35( ) and adjust to 18.5 mc.

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(e) Feed in a strong signal in from the first input generator and retune this generator to give zero beat on the frequency meter. Turn output of the generator to zero.
(f) Set frequency meter to 18.4 mc. and feed in a strong signal from the second generator, retune this generator to give zero beat on the frequency meter. Turn output of this generator to zero.
(g) Disconnect the frequency meter and increase output of both generators so that the microvolt output of both generators are the same until a 10 db signal plus noise-to-noise ratio is obtained across a standard output load. Adjust the frequency of the second generator slightly until maximum output reading is obtained. If necessary decrease both generator outputs until 10 db is obtained. Record microvolt input.
(h) Repeat steps (d) through (g) above, setting frequency meter to 18.4 and 18.2 for the second measurement, 18.7 and 18.8 for the third measurement, and 18.8 and 19.0 for the fourth measurement.
(i) Repeat steps (d) through (h), interchanging frequencies of the first and second generators.

4.5.6 Cross modulation.-

4.5.6.1 Test set up.- The test set up shall be as specified for intermodulation in 4.5.5.1.

4.5.6.2 Method of test.-

(a) Same as (a) through (c) of 4.5.5.2.
(b) Connect frequency meter to scan channel of AN/URR-35( ) and set meter to 18.5 mc. and feed a strong signal in from the first generator and adjust its frequency to give zero beat on the frequency meter. Turn output of generator to zero and disconnect frequency meter.
(c) Set output of second generator to reference microvolts obtained in (c) of 4.5.5.2. Increase output of first generator until voltmeter reads 1.2 volts. Record microvolts of first generator.
(d) Repeat (c) above with first generator modulated 90 percent.
(e) Connect frequency meter as in (b) above and adjust it to 18.4 mc. repeating step (b) at this frequency.
(f) Repeat steps (c) and (d) above.
(g) Repeat step (b) above at 18.2 mc.
(h) Repeat steps (c) and (d) above.
(i) Repeat step (b) above at 18.7 mc.
(j) Repeat steps (c) and (d) above.

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(k) Repeat step (b) above at 18.8 mc.
(l) Repeat steps (c) and (d) above.
(m) Repeat step (b) above at 19.0 mc.
(n) Repeat steps (c) and (d) above.

4.5.7 Desensitization. -

4.5.7.1 Test set up. - The test set up shall be as specified for intermodulation in 4.5.5.1.

4.5.7.2 Method of test. -

(a) Same as (a) through (d) of 4.5.5.2.
(b) Feed a strong signal from second generator and adjust its frequency to give zero beat. Disconnect frequency meter.
(c) Adjust first generator same as in step (b) of 4.5.5.2 and increase output of second generator until the output of the AN/URR-35( ) falls 2 db from 1.9 volts. Record microvolts of second generator.
(d) Connect frequency meter to scan channel of AN/URR-35( ) and adjust to 18.4 mc.
(e) Repeat steps (b) and (c) above.
(f) Repeat step (d) above at 18.3 mc.
(g) Repeat steps (b) and (c) above.
(h) Repeat step (d) above at 18.7 mc.
(i) Repeat steps (b) and (c) above.
(j) Repeat step (d) above at 18.8 mc.
(k) Repeat steps (b) and (c) above.
(l) Repeat step (d) above at 18.9 mc.
(m) Repeat steps (b) and (c) above.

4.5.8 Image rejection. -

4.5.8.1 Test set up. -

(a) Connect input generator, capable of generating frequencies between 225 and 450 mc., through a 6 db pad and a 3-foot length of 52-ohm coaxial cable to the AN/URR-35( ) input.
(b) Connect a d.c. electron tube voltmeter to a.g.c. test jack of the AN/URR-35( ).

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4.5.8.2 Method of test.-

(a) Adjust the AN/URR-35( ) to the desired frequency with the oscillator crystal controlled.
(b) Adjust input generator to the resonant frequency of the AN/URR-35( ) and record the a.g.c. voltage output with a 2-microvolt input.
(c) Adjust generator to a frequency approximately 37 mc. above the signal frequency and record the input required to obtain the same a.g.c. voltage obtained in (b) above.
(d) Repeat (c) above with a signal 3.55 mc. above the signal frequency.
(e) Repeat (b) through (d) above with AN/URR-35( ) operating with self-excited oscillator.

4.5.9 I.f. rejection.-

4.5.9.1 Test set up.- The test set up shall be as specified for image rejection in 4.5.8.1.

4.5.9.2 Test method.-

(a) Same as (a) and (b) of 4.5.8.2.
(b) Connect an i.f. generator, capable of generating 18.6 mc., to the antenna input of the AN/URR-35( ) through a series resistor selected to bring the impednance to 50-ohms.
(c) Adjust output of i.f. generator to obtain same voltage obtained in step (b) of 4.5.8.2.
(d) Repeat steps (b) and (c) above at 1.7 mc.
(e) Repeat steps (b) through (c) above with AN/URR-35( ) operating self-excited.

4.5.10 Spurious rejection.-

4.5.10.1 Test set up.-

(a) Connect input generator, capable of generating frequencies between 100 and 1000 mc., as in (a) of 4.5.8.1.
(b) Same as (b) of 4.5.8.1.

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4.5.10.2 Test method. -

(a) Same as (a) and (b) of 4.5.8.2.
(b) Slowly sweep the band of frequencies from 100 to 1000 mc. with maximum signal output from the generator. At any point of response, determine the input required to obtain the same a.g.c. voltage as obtained in (b) of 4.5.8.2.

4.5.11 Automatic gain control characteristics. -

4.5.11.1 Test set up. - The test set up shall be as specified in 4.5.1.1.

4.5.11.2 Method of test. -

(a) Adjust the AN/URR-35( ) to the desired frequency and adjust to standard gain with 30 percent modulated r.f. input.
(b) Record the audio output at 1, 2, 3, 5, 10, 30, 100, 300, 1,000, 3,000, 10,000, 30,000, and 100,000 microvolts input.

4.5.12 Automatic gain control time constant. -

4.5.12.1 Test set up. - Connect the equipment as shown on figure 1.

4.5.12.2 Method of test. -

(a) Adjust the AN/URR-35( ) to the frequency where test will be made.
(b) Connect the oscilloscope to the diode connections of the AN/URR-35( ) and determine time constant with input signals of 10,000 and 50,000 microvolts.

4.5.13 A.f. level control characteristics. -

4.5.13.1 Test set up. -

(a) Connect a signal generator, capable of generating 1.775 mc., into the grid of the second 1.775 mc. i.f. amplifier of the AN/URR-35( ).

4.5.13.2 Method of test. -

(a) Adjust the a.f. level control of the AN/URR-35( ) for maximum output.
(b) Adjust signal input until 6 milliwatt output level is obtained.
(c) Record attenuation in db below 6 milliwatts vs level control angular position as follows:
Max., 270, 240, 210, 180, 150, 120, 90, 60, 30, 0.

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4.5.14 Noise peak limiter characteristics.-

4.5.14.1 Test set up.- The test set up shall be as specified in 4.5.5.1.

4.5.14.2 Method of test.-

(a) Same as (a) through (c) of 4.5.5.2.
(b) Turn noise limiter switch of AN/URR-35( ) on.
(c) Connect pulse generator to second generator and set first generator to 2 times the value required for standard signal plus noise-to-noise ratio.
(d) Use of pulse repetition rate of 500 and a 10 microsecond pulse length, increase the output of the second generator until the signal plus noise-to-noise ratio deteriorates to 10 db.

4.5.15 Silencer control characteristics.-

4.5.15.1 Test set up.-

(a) Same as specified for image in 4.5.8.1.
(b) Terminate the AN/URR-35( ) into a 600-ohm load and connect across this resistor a voltmeter.

4.5.15.2 Method of test.-

(a) Adjust the receiver to the desired frequency, noise limiter "ON", silencer "ON", and gain to standard gain.
(b) Record the a.f. output vs r.f. input for the silencer control set with no signal input at the point where noise if reduced 35 db below standard audio output.
(c) Repeat (b) above with 80 percent rotation of the silencer control.
(d) Record signal input required at maximum setting of silencer control which will open the silencer.

4.5.16 Regeneration and instability.-

4.5.16.1 Test set up.- Install a complete set of tubes having high GM values and check operation of AN/URR-35( ) with a.g.c. on.

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4.5.16.2 Method of test.-

(a) Tune throughout the range of the AN/URR-35( ) and record frequency and output reading of any squeals.
(b) Measure sensitivity and reserve gain at low, middle, and high end of range per test specified in 4.5.1 and 4.5.2.
(c) Measure i.f. selectivity as specified in 4.5.4.1 and 4.5.4.2.

4.5.17 Line filter characteristics.-

4.5.17.1 Test set up.- Adjust receiver to standard sensitivity as specified in 4.5.1.1 and 4.5.1.2.

4.5.17.2 Method of test.-

(a) Feed full output signal generator modulated 30 percent at 1000 cycles through source impedance of 50-ohm to each leg of the power input and tune generator from 1 to 400 mc. and record any responses.
(b) Repeat (a) above with generator connected to each leg of the audio output.
(c) The attenuation in db is calculated from the ratio of input microvolts required on the terminals to the input microvolts required at the antenna.

4.5.18 Overall fidelity.-

4.5.18.1 Test set up.-

(a) Connect a signal generator, capable of generating 18.6 mc., to the jack in the output of the first mixer of the AN/URR-35( ) Modulate the signal generator 30 percent with an external audio oscillator.
(b) Connect a voltmeter across the standard output load of the AN/URR-35( ).

4.5.18.2 Method of test.-

(a) Adjust the output of the signal generator to 20,000 microvolts.
(b) Adjust the audio level control of the AN/URR-35( ) to give 1.9 volts on the output meter with 1000 cycle modulation.
(c) Keeping the modulation at 30 percent, vary the frequency of modulation from 30 to 10,000 c.p.s., recording db above and below 1000 cycles at 30, 100, 20, 350, 600, 1000, 2000, 3500, 5000, 6000, 7000 and 10,000 c.p.s. and any other frequencies needed to plot an overall fidelity curve.

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4.5.19 Frequency and gain stability. -

4.5.19.1 Line voltage variations. -

4.5.19.1.1 Method of test. -

(a) Tune receiver with 115-volt primary input. Measure manual oscillator frequency, i.f. center frequency and reserve gain.
(b) Reduce line voltage to 103.5 volts and repeat (a) above.
(c) Inject frequency of 243 mc. into antenna of the AN/URR-35( ) and measure sensitivity.
(d) Increase line voltage to 126.5 volts and repeat (a) and (c) above.
(e) Repeat (b), (c) and (d) above at 396.6 mc.

4.5.19.2 Temperature variations. -

4.5.19.2.1 Method of test. -

(a) With the AN/URR-35( ) installed in a temperature chamber, and with power on, bring temperature to 25°C. and stabilize.
(b) Measure and record i.f. center frequency, manual oscillator frequency, reserve gain and sensitivity.
(c) Conduct test as specified in Specification MIL-E-16400 for shipboard equipment repeating (b) above at each increment of temperature change.

4.5.19.3 Humidity. -

4.5.19.3.1 Method of test. - Conduct test as specified by Specification MIL-E-16400 repeating measurements as specified in (b) of 4.5.19.2.1.

4.5.20 Maximum undistorted output. -

4.5.20.1 Test set up. -

(a) Connect generator, set to 18.6 mc., to output of the first mixer of the AN/URR-35( ) and modulate generator 30 percent at 1000 c.p.s.
(b) Connect noise and distortion meter across the standard output load of the AN/URR-35( ).

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4.5.20.2 Method of test.-

a) Adjust the output of the generator to 100,000 microvolts and adjust the AN/URR-35( ) gain to produce 6 volts across the standard output load.
(b) Measure total harmonic distortion in the audio output signal.

4.5.21 Output hum level.-

4.5.21.1 Test set-up.-

(a) Connect an input generator, capable of producing frequencies between 225 to 400 mc., to the input of the AN/URR-35( ) through a 6 db pad and a 3-foot length of 52-ohm coaxial cable.
(b) Connect a generator, adjusted to 18.6 mc. to the output of the first mixer of the AN/URR-35( ).
(c) Connect a voltmeter across the standard output load of the AN/URR-35( ).
(d) Connect d.c. vacuum tube voltmeter to a.g.c. of AN/URR-35( ).

4.5.21.2 Method of test.-

(a) Turn audio level control of the AN/URR-35( ) to maximum and short the grid of the final i.f. amplifier to ground and record reading on output voltmeter.
(b) Adjust AN/URR-35( ) and input generator to the desired frequency, adjust output of generator to 10,000 microvolts and record a.g.c. voltage output of the AN/URR-35( ). Reduce output of input generator to zero.
(c) Modulate i.f. generator 30 percent at 1000 c.p.s. and adjust output of generator to obtain same a.g.c. voltage recorded in (b) above.
(d) Adjust audio level control of AN/URR-35( ) until output voltmeter reads 1.9 volts. Reduce output of i.f. generator to zero.
(e) Record readings on output voltmeter with the input generator adjusted for 100, 500, 1000, 5000, 10,000 and 50,000 microvolts. Plot results in terms of db below 1.9 volts.

4.6 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.-

5.1.1 Equipment.- Each radio receiving set and technical publications shall be prepared for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment, as specified (see 6.1), in accordance with Specification MIL-E-17555.

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5.1.2 Maintenance parts. - Equipment and stock maintenance parts shall be prepared for domestic shipment and storage; or overseas shipment, as specified (see 6.1), in accordance with Specification MIL-P-17652.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Where acceptability tests will be made (see 4.2).
(c) Whether delivery shall be for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment (see 5.1.1 and 5.1.2).

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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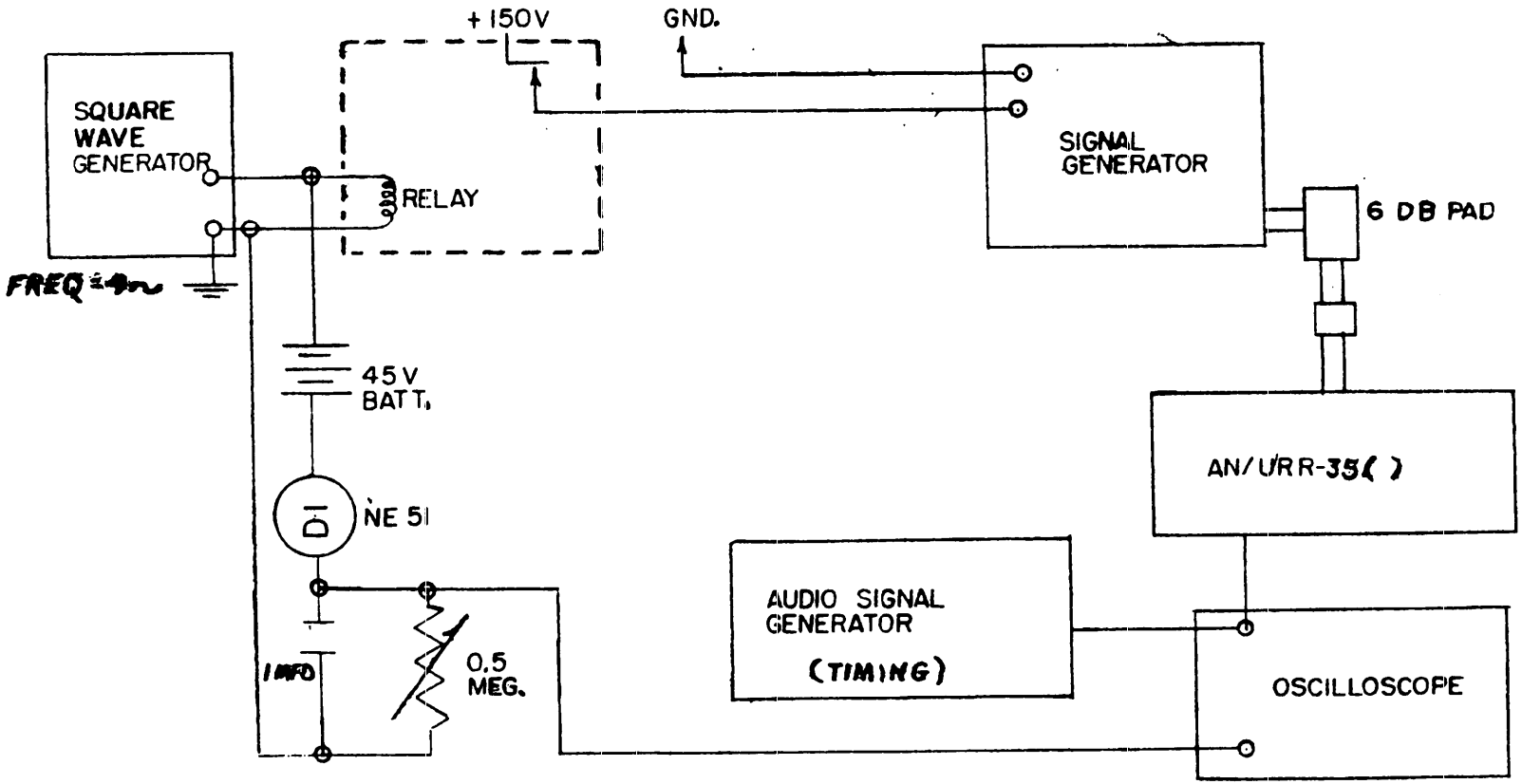


FIGURE 1.–TEST SET-UP FOR A.G.C. TIME CONSTANT

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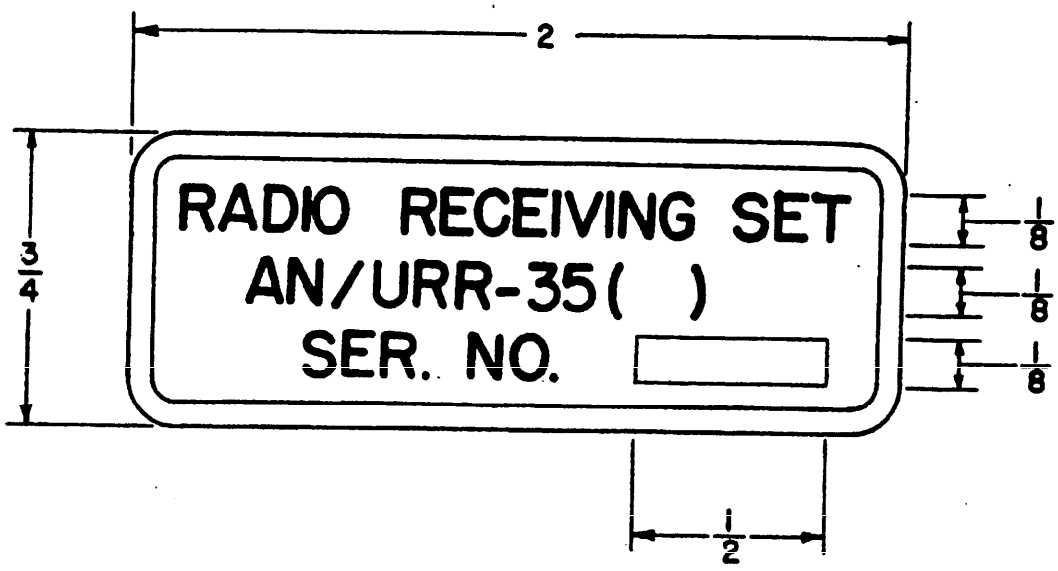


Figure 2.- Nameplate

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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